1. Which protein polymerizes to form microfilaments in eukaryotic cells?
- **(A)** Tubulin
- **(B)** Actin
- **(C)** Vimentin
- **(D)** Keratin

2. What is the diameter of an individual microtubule filament?
- **(A)** 7 nm
- **(B)** 10 nm
- **(C)** 25 nm
- **(D)** 100 nm

3. Which nucleotide binds to beta-tubulin to regulate microtubule polymerization?
- **(A)** ATP
- **(B)** GTP
- **(C)** CTP
- **(D)** UTP

4. Which motor protein moves primarily toward the plus-end of microtubules?
- **(A)** Cytoplasmic dynein
- **(B)** Kinesin
- **(C)** Myosin II
- **(D)** Myosin V

5. Which motor protein moves retrograde toward the minus-end of microtubules?
- **(A)** Kinesin-1
- **(B)** Cytoplasmic dynein
- **(C)** Myosin I
- **(D)** Kinesin-5

6. What is the primary nucleating center for microtubules in animal cells?
- **(A)** Centrosome
- **(B)** Kinetochore
- **(C)** Basal lamina
- **(D)** Nucleolus

7. Which drug binds to tubulin heterodimers and prevents microtubule polymerization?
- **(A)** Taxol
- **(B)** Colchicine
- **(C)** Phalloidin
- **(D)** Cytochalasin D

8. Which anticancer drug stabilizes microtubules against depolymerization?
- **(A)** Paclitaxel (Taxol)
- **(B)** Vinblastine
- **(C)** Nocodazole
- **(D)** Latrunculin

9. Which actin-binding toxin binds and stabilizes F-actin filaments?
- **(A)** Cytochalasin B
- **(B)** Phalloidin
- **(C)** Latrunculin A
- **(D)** Swinholide

10. Intermediate filaments have an approximate diameter of:
- **(A)** 7 nm
- **(B)** 10-12 nm
- **(C)** 25 nm
- **(D)** 50 nm

11. Which intermediate filament protein is characteristic of epithelial cells?
- **(A)** Desmin
- **(B)** Vimentin
- **(C)** Keratin
- **(D)** Neurofilament

12. Which intermediate filament is expressed in mesenchymal cells like fibroblasts?
- **(A)** Vimentin
- **(B)** Keratin
- **(C)** Glial fibrillary acidic protein
- **(D)** Desmin

13. Vesicles moving anterograde from the rough ER to the cis-Golgi are coated with:
- **(A)** Clathrin
- **(B)** COP I
- **(C)** COP II
- **(D)** Caveolin

14. Retrograde vesicular transport from Golgi back to ER is mediated by:
- **(A)** COP II
- **(B)** COP I
- **(C)** Clathrin
- **(D)** Retromer

15. The KDEL receptor retrieves ER-resident proteins from Golgi by recognizing:

(A) KDEL sequence at C-terminus
(B) N-terminal signal peptide
(C) Nuclear localization signal
(D) Mannose-6-phosphate

16. Lysosomal acid hydrolases are tagged for targeting in the cis-Golgi by addition of:

(A) Glucose-6-phosphate
(B) Mannose-6-phosphate (M6P)
(C) Galactose
(D) Sialic acid

17. Which small GTPase regulates the assembly of COP II coats at the ER membrane?

(A) Sar1
(B) Arf1
(C) Ran
(D) Rac1

18. Which small GTPase regulates COP I and clathrin coat assembly at the Golgi?

(A) Arf1
(B) Sar1
(C) Cdc42
(D) RhoA

19. The 9+2 axoneme arrangement of microtubules is characteristic of:

(A) Primary cilia
(B) Motile cilia and flagella
(C) Centrioles
(D) Basal bodies

20. The basal body and centriole share which structural microtubule arrangement?

(A) 9+0 doublet
(B) 9+2 doublet
(C) 9+0 triplet
(D) 13 singlet

21. Which motor protein generates sliding forces between outer doublets in motile cilia?

(A) Axonemal dynein
(B) Cytoplasmic dynein
(C) Kinesin-2
(D) Myosin VII

22. Actin treadmilling occurs when G-actin addition at the plus-end equals:

(A) Addition at minus-end
(B) Dissociation at minus-end
(C) ATP synthesis rate
(D) Capping protein binding

23. Which actin-binding protein caps the plus-barbed end of F-actin filaments?

(A) CapZ
(B) Tropomodulin
(C) Profilin
(D) Cofilin

24. Which protein severs older ADP-actin filaments near the pointed minus-end?

(A) Gelsolin
(B) Cofilin (ADF)
(C) Formin
(D) Fimbrin

25. Which nucleation promoting factor drives branched actin network formation at the leading edge?

(A) Arp2/3 complex
(B) Formin
(C) Spire
(D) Ena/VASP

26. Formins nucleate and processively elongate which type of actin structures?

(A) Branched meshworks
(B) Unbranched linear actin bundles
(C) Cortical rings
(D) Microvilli cores

27. Which Rho-family GTPase promotes lamellipodia formation via Arp2/3 activation?

(A) RhoA
(B) Rac1
(C) Cdc42
(D) Rab5

28. Which Rho-family GTPase promotes filopodia formation via WASP and formins?

(A) Cdc42
(B) RhoA
(C) Rac1
(D) Ras

29. Activation of RhoA promotes the formation of which cytoskeletal structures?
- **(A)** Filopodia
- **(B)** Lamellipodia
- **(C)** Actin stress fibers and focal adhesions
- **(D)** Primary cilia

30. Myosin II is the two-headed conventional motor responsible for:
- **(A)** Muscle contraction and cytokinesis
- **(B)** Organelle transport
- **(C)** Endocytosis
- **(D)** Auditory hair cell tension

31. Myosin V is specialized for processive vesicle transport because of its:
- **(A)** High ATPase speed
- **(B)** Long neck domains and high duty ratio
- **(C)** Monomeric structure
- **(D)** Inability to bind actin

32. Which enzyme phosphorylates the regulatory light chain of myosin II to activate it?
- **(A)** Myosin light chain kinase (MLCK)
- **(B)** Protein kinase A
- **(C)** MAP kinase
- **(D)** Src kinase

33. Calcium binds to which regulatory subunit to trigger skeletal muscle contraction?
- **(A)** Tropomyosin
- **(B)** Troponin C
- **(C)** Troponin I
- **(D)** Troponin T

34. In smooth muscle, Ca2+ activates contraction by first binding to:
- **(A)** Calmodulin
- **(B)** Troponin C
- **(C)** Parvalbumin
- **(D)** Calsequestrin

35. The primary calcium release channel in the sarcoplasmic reticulum membrane is:
- **(A)** Ryanodine receptor (RyR)
- **(B)** IP3 receptor
- **(C)** Voltage-gated calcium channel
- **(D)** NCX antiporter

36. Which organelle functions as the central sorting hub for endocytosed macromolecules?
- **(A)** Late endosome
- **(B)** Early endosome
- **(C)** Trans-Golgi network
- **(D)** Lysosome

37. Caveolae are flask-shaped plasma membrane invaginations rich in:
- **(A)** Caveolin and cholesterol
- **(B)** Clathrin and adaptor proteins
- **(C)** COP I
- **(D)** Spectrin

38. The protein coat that retrieves unoccupied M6P receptors from endosomes to Golgi is:
- **(A)** Retromer
- **(B)** COP I
- **(C)** COP II
- **(D)** Clathrin

39. Multivesicular bodies (MVBs) are formed by inward budding mediated by:
- **(A)** ESCRT complexes
- **(B)** SNARE proteins
- **(C)** Dynamin
- **(D)** Spg1

40. Ubiquitination of membrane receptors targets them for internalization and degradation by:
- **(A)** Proteasome
- **(B)** Lysosome via ESCRT
- **(C)** Peroxisome
- **(D)** Mitochondria

41. Dynamic instability of microtubules is characterized by stochastic switching between:
- **(A)** Phosphorylation and dephosphorylation
- **(B)** Growing (polymerization) and shrinking (catastrophe)
- **(C)** Sliding and folding
- **(D)** Branching and severing

42. The structural basis of microtubule catastrophe is the loss of the:
- **(A)** GTP cap at the plus-end
- **(B)** GDP core at the minus-end
- **(C)** Gamma-tubulin ring
- **(D)** Kinesin motor

43. The gamma-tubulin ring complex (gamma-TuRC) nucleates microtubule assembly at the:
- **(A)** Minus-end at the centrosome
- **(B)** Plus-end in the cytoplasm
- **(C)** Kinetochore
- **(D)** Axoneme tip

44. Which plus-end tracking protein (+TIP) tracks growing microtubule ends and recruits partners?
- **(A)** EB1
- **(B)** MCAK
- **(C)** Stathmin
- **(D)** Katanin

45. Stathmin (oncoprotein 18) promotes microtubule depolymerization by:
- **(A)** Severing microtubules
- **(B)** Sequestering two tubulin heterodimers
- **(C)** Capping plus-ends
- **(D)** Phosphorylating beta-tubulin

46. Katanin is an AAA+ ATPase specialized for:
- **(A)** Microtubule nucleation
- **(B)** Microtubule severing
- **(C)** Actin capping
- **(D)** Vesicle scission

47. Tau and MAP2 stabilize microtubules by binding along the protofilament lattice in:
- **(A)** Axons and dendrites respectively
- **(B)** Dendrites and axons respectively
- **(C)** Muscle sarcomeres
- **(D)** Epithelial microvilli

48. Hyperphosphorylation of Tau leads to neurofibrillary tangle formation in:
- **(A)** Parkinson disease
- **(B)** Alzheimer disease
- **(C)** Huntington disease
- **(D)** Amyotrophic lateral sclerosis

49. Profilin promotes actin polymerization by catalyzing:
- **(A)** ADP-to-ATP nucleotide exchange on G-actin
- **(B)** G-actin phosphorylation
- **(C)** Actin filament severing
- **(D)** Pointed-end capping

50. Thymosin beta-4 maintains a soluble monomeric actin pool by:
- **(A)** Sequestering ATP-G-actin to prevent spontaneous nucleation
- **(B)** Phosphorylating G-actin
- **(C)** Nucleating branched actin
- **(D)** Capping barbed ends

51. WASP (Wiskott-Aldrich Syndrome Protein) activates Arp2/3 upon binding to:
- **(A)** Active Cdc42-GTP and PI(4,5)P2
- **(B)** Active RhoA-GTP
- **(C)** Inactive Rab-GDP
- **(D)** ATP

52. During muscle contraction, the power stroke occurs immediately following:
- **(A)** ATP binding to myosin head
- **(B)** Release of inorganic phosphate (Pi)
- **(C)** ATP hydrolysis to ADP+Pi
- **(D)** Binding of ADP

53. Rigor mortis occurs after death because lack of ATP prevents:
- **(A)** Myosin detachment from the actin filament
- **(B)** Myosin power stroke
- **(C)** Calcium release from SR
- **(D)** Troponin C activation

54. The step size of conventional kinesin-1 moving along a protofilament is:
- **(A)** 4 nm
- **(B)** 8 nm (tubulin dimer repeat)
- **(C)** 16 nm
- **(D)** 36 nm

55. Kinesin-1 moves processively via a mechanism known as:
- **(A)** Inchworm model
- **(B)** Hand-over-hand walking
- **(C)** Diffusive hopping
- **(D)** Power-stroke rolling

56. Cytoplasmic dynein requires which large multi-subunit complex to link to cargo and enhance processivity?
- **(A)** Dynactin
- **(B)** Kinesin-light chain
- **(C)** Adaptin
- **(D)** Spectrin

57. Lis1 and Nde1 regulate dynein motor function by directly binding to the:

(A) Dynein motor AAA+ ring
(B) Dynactin p150-Glued
(C) Microtubule minus-end
(D) Golgi membrane

58. In Kartagener syndrome (primary ciliary dyskinesia), genetic mutations disrupt:

(A) Axonemal dynein arms
(B) Cytoplasmic kinesin
(C) Tubulin synthesis
(D) Keratin filaments

59. The triad junction in skeletal muscle consists of a central T-tubule flanked by two:

(A) Terminal cisternae of sarcoplasmic reticulum
(B) Mitochondria
(C) Golgi stacks
(D) Lysosomes

60. The voltage sensor in the skeletal muscle T-tubule membrane is the:

(A) Dihydropyridine receptor (DHPR / Cav1.1)
(B) Ryanodine receptor (RyR1)
(C) SERCA pump
(D) Calsequestrin

61. In skeletal muscle, DHPR is mechanically coupled directly to:

(A) RyR1 on the SR membrane
(B) RyR2
(C) IP3 receptor
(D) Troponin T

62. In cardiac muscle, calcium influx through Cav1.2 triggers calcium release via:

(A) Mechanical coupling
(B) Calcium-Induced Calcium Release (CICR) via RyR2
(C) IP3 generation
(D) Calsequestrin phosphorylation

63. Sec23/Sec24 heterodimer in the COP II coat serves which specific role?

(A) Sec24 captures cargo receptors, Sec23 acts as GAP for Sar1
(B) Sec23 binds GTP
(C) Sec24 cleaves lipids
(D) Sec23 forms the outer cage

64. The outer scaffolding cage of the COP II coat is assembled from:

(A) Sec13/Sec31 heterotetramers
(B) Sec23/Sec24
(C) Clathrin triskelions
(D) Coatomer

65. Uncoating of COP II vesicles is triggered by:

(A) GTP hydrolysis by Sar1 promoted by Sec23
(B) ATP hydrolysis by NSF
(C) Phosphorylation by PKA
(D) Calcium binding

66. The KDEL receptor has high ligand affinity in the Golgi and low affinity in the ER because:

(A) Golgi lumen is more acidic (pH 6.2) than ER lumen (pH 7.2)
(B) ER is more acidic than Golgi
(C) Golgi has higher ATP concentration
(D) ER lacks calcium

67. I-cell disease (Mucolipidosis II) is caused by a congenital deficiency of:

(A) GlcNAc phosphotransferase
(B) Acid phosphatase
(C) Cathepsin D
(D) Hexosaminidase A

68. In I-cell disease, lysosomal enzymes lack M6P tags and are abnormally:

(A) Retained in the ER
(B) Secreted constitutively into extracellular fluids
(C) Degraded in the proteasome
(D) Imported into mitochondria

69. Which Rab GTPase marks early endosomes and recruits EEA1 tethering protein?

(A) Rab1
(B) Rab5
(C) Rab7
(D) Rab11

70. The endosomal maturation transition from early to late endosomes involves a Rab conversion from:

(A) Rab5 to Rab7
(B) Rab7 to Rab5
(C) Rab1 to Rab6
(D) Rab4 to Rab11

71. Which recycling endosome Rab directs recycling of transferrin receptors back to the plasma membrane?

(A) Rab11
(B) Rab7
(C) Rab2
(D) Rab9

72. The ESCRT-0 complex recognizes ubiquitinated cargo on endosomal membranes via its:

(A) HRS and STAM subunits containing UIM motifs
(B) SNARE motifs
(C) AAA+ ATPase domain
(D) Clathrin-box

73. ESCRT-III polymers form spiral filaments that drive:

(A) Membrane scission with neck constriction away from cytoplasm
(B) Clathrin uncoating
(C) COP II assembly
(D) Vesicle tethering

74. Vps4 is an AAA+ ATPase required for the ATP-dependent disassembly of:

(A) ESCRT-III filaments
(B) SNARE complexes
(C) Clathrin cages
(D) Dynactin

75. Botulinum neurotoxins A and E cleave which specific SNARE protein to paralyze neurotransmission?

(A) SNAP-25
(B) Syntaxin-1
(C) Synaptobrevin (VAMP2)
(D) Synaptotagmin

76. Tetanus toxin and Botulinum toxins B, D, F, and G specifically cleave which SNARE component?

(A) Synaptobrevin / VAMP
(B) SNAP-25
(C) Syntaxin-1
(D) Complexin

77. Synaptotagmin-1 functions as the primary calcium sensor for synaptic vesicle fusion via its:

(A) Tandem C2A and C2B domains
(B) SNARE motif
(C) Transmembrane helix
(D) PH domain

78. Complexin stabilizes primed trans-SNARE complexes in a clamped, fusion-ready intermediate until:

(A) Ca2+ binds to synaptotagmin to relieve the clamp
(B) ATP is hydrolyzed
(C) Dynamin pinches the membrane
(D) pH drops

79. The stall force of a single kinesin-1 motor molecule measured by optical tweezers is approximately:

(A) 1-2 pN
(B) 5-7 pN
(C) 20-25 pN
(D) 100 pN

80. The stall force of a single skeletal muscle myosin II motor head is approximately:

(A) 3-5 pN
(B) 10-15 pN
(C) 50 pN
(D) 0.5 pN

81. Myosin V takes 36 nm hand-over-hand steps along actin filaments, which corresponds to the:

(A) Pseudo-repeat helical pitch of the actin filament (36 nm)
(B) Tubulin dimer length
(C) Distance between Z-discs
(D) Sperm flagellum wavelength

82. Optical trap interferometry reveals that kinesin-1 hydrolyzes how many ATP molecules per 8 nm step?

(A) Exactly 1 ATP per 8 nm step
(B) 2 ATP per step
(C) 4 ATP per step
(D) Variable ATP based on load

83. What is the critical concentration (Cc) relationship governing actin treadmilling at steady state?

(A) Cc(minus) > C_steady > Cc(plus)
(B) Cc(plus) > C_steady > Cc(minus)
(C) Cc(plus) = Cc(minus)
(D) C_steady = 0

84. At 25 deg C with physiological salts, the critical concentration of G-actin at the barbed (+) end is ~0.1 microM, while at the pointed (-) end it is:

(A) 0.1 microM
(B) 0.6 to 0.8 microM
(C) 10 microM
(D) 0.01 microM

85. In the sliding filament model, during sarcomere shortening, which band remains constant in length?

(A) I-band
(B) H-zone
(C) A-band
(D) Sarcomere length

86. Titin is a giant sarcomeric protein (approx 3.8 MDa) that connects the:

(A) Z-disc to the M-line, acting as a molecular spring
(B) Z-disc to actin
(C) M-line to myosin alone
(D) T-tubule to SR

87. Nebulin is an inelastic ruler protein that determines the precise length of:

(A) Actin thin filaments in skeletal muscle
(B) Thick filaments
(C) Microtubules
(D) T-tubules

88. The Arp2/3 complex binds to the side of an existing actin mother filament to nucleate a daughter branch at an angle of:

(A) 45 degrees
(B) 70 degrees
(C) 90 degrees
(D) 120 degrees

89. Listeria monocytogenes hijacks host cell actin polymerization for intracellular propulsion using its surface protein:

(A) ActA (which mimics WASP to activate Arp2/3)
(B) InlA
(C) Listeriolysin O
(D) Internalin B

90. In the primary cilium, the BBSome complex functions as an adaptor machine for:

(A) Ciliary membrane cargo trafficking
(B) Tubulin cleavage
(C) Dynein power stroke
(D) Flagellar rotation

91. Mutations in Bardet-Biedl syndrome (BBS) genes cause ciliopathies characterized by:

(A) Retinal degeneration, obesity, polydactyly, renal cysts
(B) Hypertrophic cardiomyopathy alone
(C) Severe skin blistering
(D) Erythrocyte fragility

92. Kinesin-13 family members (e.g., MCAK) do not perform processive transport; instead, they function as:

(A) ATP-dependent microtubule end depolymerases
(B) Microtubule nucleators
(C) Actin bundle crosslinkers
(D) Vesicle tethers

93. Kinesin-14 family members (e.g., Ncd) possess a catalytic motor domain at their C-terminus and move:

(A) Retrograde toward the microtubule minus-end
(B) Anterograde toward plus-end
(C) Bidirectionally
(D) Randomly

94. During mitosis, Kinesin-5 (Eg5 / BimC) is a bipolar homotetrameric motor that:

(A) Crosslinks and slides antiparallel polar microtubules apart to separate spindle poles
(B) Depolymerizes kinetochore fibers
(C) Translocates chromosomes to centrosomes
(D) Forms the contractile ring

95. Blebbistatin is a cell-permeable small molecule that specifically inhibits:

(A) Myosin II ATPase by trapping it in an actin-detached state
(B) Kinesin-1
(C) Dynein
(D) Actin polymerization

96. Monastrol is an allosteric inhibitor of Kinesin-5 (Eg5) that arrests dividing cells in:

(A) Monoastral spindle state during mitosis
(B) G1 phase
(C) Anaphase B
(D) Cytokinesis

97. The intermediate filament dimer forms a coiled-coil between two parallel alpha-helices, which then assemble into:

(A) Antiparallel, staggered tetramers lacking structural polarity
(B) Polar trimers
(C) Helical doublets
(D) Head-to-tail polymers

98. Because intermediate filament tetramers are non-polar, motor proteins (kinesin/myosin):

(A) Cannot utilize intermediate filaments as tracks for directed transport
(B) Move at double velocity
(C) Only move toward the nucleus
(D) Form branches

99. Epidermolysis Bullosa Simplex (EBS) is caused by dominant-negative missense mutations in:

(A) Keratin 5 or Keratin 14
(B) Keratin 1 or 10
(C) Desmin
(D) Lamin A

100. Desmin-related myopathy is a debilitating cardiac and skeletal muscle pathology caused by mutations in the:

(A) Desmin gene, disrupting intermediate filament networks at Z-discs
(B) Dystrophin gene
(C) Myosin heavy chain
(D) Actin gene

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Contrast the structural architectures, monomer subunits, nucleotide dependencies, and polarities of Microfilaments, Microtubules, and Intermediate Filaments.

Q2. Explain the molecular mechanism of Microtubule Dynamic Instability and detail the structural role of the GTP cap.

Q3. Describe the hand-over-hand processive motility mechanism of Kinesin-1 along a microtubule protofilament.

Q4. Explain the molecular mechanism of Actin Treadmilling, deriving the steady-state conditions using critical concentrations.

Q5. Describe the Arp2/3 complex nucleation mechanism and explain how it drives branched actin networks at lamellipodia.

Q6. Explain the cross-bridge cycle of Skeletal Muscle Myosin II and detail the structural triggers of the power stroke.

Q7. Discuss the structural organization of Sarcomeres and detail the regulatory roles of the Troponin-Tropomyosin complex in excitation-contraction coupling.

Q8. Detail the molecular mechanism of COP II coat assembly, cargo selection, and vesicle scission at the Endoplasmic Reticulum.

Q9. Explain the retrieval mechanism of ER-resident proteins from the Golgi via the KDEL receptor and COP I retrograde transport.

Q10. Describe the Mannose-6-Phosphate (M6P) pathway for lysosomal enzyme targeting and detail the biochemical defect in I-Cell Disease.

Q11. Explain the structural mechanics of Clathrin-coated vesicle formation, contrasting the roles of AP-2 adaptors, Clathrin triskelions, and Dynamin GTPase.

Q12. Detail the core components and biochemical steps of the ESCRT machinery in multivesicular body (MVB) sorting.

Q13. Discuss the molecular architecture of the 9+2 Axoneme in motile cilia and explain how sliding of outer doublets is converted into bending.

Q14. Explain the pathophysiology of Primary Ciliary Dyskinesia (Kartagener Syndrome) and its link to situs inversus.

Q15. Describe the biochemical structure, assembly, and tissue distribution of Intermediate Filaments, contrasting acidic/basic keratins, vimentin, desmin, and neurofilaments.

Q16. Explain the molecular pathogenesis of Epidermolysis Bullosa Simplex (EBS) and its relationship to Keratin 5 and Keratin 14 mutations.

Q17. Discuss the structural organization and functional role of Focal Adhesions as mechanosensory hubs connecting the extracellular matrix to actin stress fibers.

Q18. Explain how the Kinesin-5 (Eg5) motor operates during mitosis to establish bipolar mitotic spindle architecture.

Q19. Describe the mechanism of vesicular transport from the Trans-Golgi Network (TGN) to the plasma membrane via Constitutive versus Regulated Secretory pathways.

Q20. Explain the functional role of the Retromer complex in endosome-to-Golgi retrograde transport and its clinical significance in Alzheimer's disease.

Q21. Detail the molecular mechanism of Caveolae formation and their physiological functions in endothelial mechanosensing and endocytosis.

Q22. Explain how bacterial pathogens like Listeria and Shigella utilize host cell actin polymerization for intracellular motility and cell-to-cell spread.

Q23. Describe the architecture and mechanochemical cycle of Cytoplasmic Dynein, and detail how Dynactin and Lis1 regulate its processivity.

Q24. Explain the structural mechanics of the Nuclear Envelope and describe how the LINC complex connects the nucleoskeleton to the cytoskeleton.

Q25. Discuss the role of Phosphoinositides (PIPs) as spatial landmark codes regulating organelle identity and vesicular trafficking.

Q26. Explain the molecular mechanism of Phagocytosis, detailing Fc-receptor recognition, pseudopod extension, and phagolysosome maturation.

Q27. Describe the biophysical properties and structural basis of the Titin molecular spring in muscle passive elasticity.

Q28. Explain the role of Septins as a fourth cytoskeletal component in forming diffusion barriers and scaffolding signaling complexes.

Q29. Discuss the molecular mechanism of Cilia Disassembly during cell cycle entry and explain why primary cilia must be resorbed before mitosis.

Q30. Explain the molecular pathology of Hutchinson-Gilford Progeria Syndrome (HGPS) and how Farnesyltransferase Inhibitors (FTIs) provide therapeutic benefit.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[B]	**Q21:**	[A]	**Q41:**	[B]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[C]	**Q22:**	[B]	**Q42:**	[A]	**Q62:**	[B]	**Q82:**	[A]
Q3:	[B]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[B]	**Q24:**	[B]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[B]
Q5:	[B]	**Q25:**	[A]	**Q45:**	[B]	**Q65:**	[A]	**Q85:**	[C]
Q6:	[A]	**Q26:**	[B]	**Q46:**	[B]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[B]	**Q27:**	[B]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[B]	**Q68:**	[B]	**Q88:**	[B]
Q9:	[B]	**Q29:**	[C]	**Q49:**	[A]	**Q69:**	[B]	**Q89:**	[A]
Q10:	[B]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[C]	**Q31:**	[B]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[B]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[C]	**Q33:**	[B]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[B]	**Q34:**	[A]	**Q54:**	[B]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[B]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[B]	**Q36:**	[B]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[B]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[B]	**Q99:**	[A]
Q20:	[C]	**Q40:**	[B]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

SECTION B: DETAILED SOLUTIONS FOR SUBJECTIVE QUESTIONS Dynamic instability and treadmilling are structural roles of Microfilaments. Microtubules, and I

- Rescue: If a shrinking end binds newly synthesized GTP-tubulin dimers or encounters an isolated GTP-island along the shaft, the GTP cap is reconstituted, halting depolymerization and resuming steady elongation.

>>Solution Question Describe the molecular mechanisms of Actin Treadmilling driven by the steady-state conditions at critical concentrations.

- At the pointed end: [G] < Cc(-), so net depolymerization occurs (subunits continuously dissociate).
- At steady state, the rate of barbed-end subunit addition exactly equals the rate of pointed-end subunit detachment:
k_on(+) * [G] - k_off(+) = k_off(-) - k_on(-) * [G].
- Result: Net filament length remains invariant while individual actin monomers tread continuously from the barbed end toward the pointed end.

>>Solution Question Bank Describe the Arp2/3 complex of Skeletal Muscle Myosin and explain in detail the structural hierarchy of the actin filaments.

The cocked myosin-ADP-Pi head binds weakly to a new actin monomer located 11 nm further along the thin filament toward the Z-disc.

5. State 5 (Phosphate Release and Power Stroke):
Release of inorganic phosphate (Pi) tightens actin binding and triggers the mechanical Power Stroke: the lever arm snaps back 70 degrees toward its original uncocked position, pulling the actin filament ~8-10 nm toward the center of the sarcomere (M-line), generating a peak force of 3-5 pN per head.

6. State 6 (ADP Release):
ADP dissociates from the active site, returning the complex to the high-affinity rigor state awaiting the next ATP molecule.

>>Solution Question Bank: Detailed step-by-step solutions for all questions Study Tools: Flashcards, mind maps, and practice tests Exam Prep: Strategies, tips, and last-minute revision guides Join Now: Click the link in bio to start your journey!

2. Inner Coat Assembly and Cargo Recruitment (Sec23/Sec24):
Membrane-anchored Sar1-GTP recruits the Sec23/Sec24 heterodimer:
- Sec24 possesses multiple cargo-binding pockets that recognize specific sorting motifs in the cytosolic tails of transmembrane cargo (e.g., di-acidic [DXE] or di-hydrophobic motifs).
- Soluble luminal cargo (e.g., procollagen) binds transmembrane cargo receptors (e.g., ERGIC-53) that present cytosolic motifs to Sec24.
- Sec23 binds directly to Sar1-GTP, functioning as its GTPase-Activating Protein (GAP).
3. Outer Cage Assembly and Budding (Sec13/Sec31):
Sec23/Sec24 complexes recruit outer shell Sec13/Sec31 heterotetramers. Sec13/Sec31 molecules polymerize into a flexible cuboctahedral or icosahedral cage that exerts mechanical force, curving the ER membrane into a spherical bud (~60-90 nm diameter).
4. Vesicle Scission and Uncoating:
Outer coat binding accelerates the GAP activity of Sec23 toward Sar1 by 10-fold. Sar1 hydrolyzes its bound GTP to GDP, retracting its amphipathic helix from the membrane. Release of Sar1 destabilizes the coat lattice, preparing the vesicle for uncoating and subsequent tethering to the cis-Golgi.

>>Solution Question Bank Disease

synthesized in the rough ER with high-mannose N-linked glycans:
- Step 1 (cis-Golgi): UDP-N-acetylglucosamine:lysosomal-enzyme N-acetylglucosamine-1-phosphotransferase (GlcNAc phosphotransferase) recognizes a specific three-dimensional 'signal patch' on the folded hydrolase. It transfers GlcNAc-1-phosphate from UDP-GlcNAc to terminal mannose residues.
- Step 2 (trans-Golgi): The 'uncovering enzyme' (phosphodiester alpha-GlcNAcase) cleaves off the terminal GlcNAc, exposing the terminal Mannose-6-Phosphate (M6P) recognition tag.
2. Sorting in the Trans-Golgi Network (TGN):
In the TGN, transmembrane M6P Receptors (MPRs: cation-dependent CD-MPR and cation-independent CI-MPR) bind M6P residues at pH 6.5. The cytosolic tails of MPRs recruit GGA adaptors and AP-1/clathrin coats, packaging hydrolases into transport vesicles targeted to late endosomes.
3. Release in Endosomes and Receptor Recycling:
Upon fusing with late endosomes, the acidic pH (~5.0 to 5.5) triggers hydrolase dissociation from MPRs. Acid phosphatases dephosphorylate the M6P tag, preventing rebinding. The retromer complex captures empty MPRs and returns them back to the TGN.
4. Molecular Pathology of I-Cell Disease (Mucolipidosis II):
Caused by autosomal recessive mutations in the GNPTAB gene encoding the catalytic subunit of GlcNAc phosphotransferase:
- Without M6P tags, newly synthesized acid hydrolases cannot bind MPRs. Instead, they are packaged into constitutive secretory vesicles and dumped into the bloodstream and extracellular fluids.
- Lysosomes become severely deficient in digestive enzymes, accumulating undigested glycosaminoglycans, lipids, and oligosaccharides as massive dense 'inclusion bodies' (I-cells), causing severe skeletal deformities, coarse facial features, and early childhood mortality.

>> Solution to Question 11 Explain the structural mechanics of Clathrin-coated vesicle formation, contrasting the roles of AP-2 adaptors, Clathrin triskelions, and D

- Auxilin recruits the heat shock cognate chaperone Hsc70, which hydrolyzes ATP to dismantle the outer clathrin cage, yielding a naked endocytic vesicle ready for endosomal fusion.

- In an intact cilium, outer doublets are physically anchored at their base to the triple-microtubule basal body and cross-linked along their length by elastic nexin bridges.
- Nexin links mechanically resist unrestricted longitudinal sliding. Consequently, shear forces generated by dynein walking are converted into lateral bending moments.
- Asymmetric, alternating activation of dynein arms on opposite sides of the axoneme produces the characteristic ciliary stroke or sinusoidal flagellar wave.

- Protofilament & Filament: Eight tetramers associate laterally into a 10 nm unit length filament (ULF). ULFs anneal end-to-end and compact radially to form the mature 10-12 nm rope-like filament.

3. Major Classes and Tissue Distributions:

- Type I (Acidic Keratins, K9-K28) & Type II (Basic Keratins, K1-K8): Expressed in epithelial tissues. Form obligate heterodimers (one Type I + one Type II) providing tensile strength to skin and mucosal linings.
- Type III (Vimentin-like):
- Vimentin: Mesenchymal cells (fibroblasts, endothelial cells, leukocytes).
- Desmin: Striated and smooth muscle cells, crosslinking myofibrillar Z-discs to the sarcolemma.
- Glial Fibrillary Acidic Protein (GFAP): Astrocytes and glial cells of the central nervous system.
- Type IV (Neurofilaments): NF-L (light), NF-M (medium), NF-H (heavy). Expressed exclusively in neurons; determine axonal caliber and conduction velocity.
- Type V (Nuclear Lamins): Lamins A, B, C lining the inner nuclear membrane.

Focal adhesions are large, dynamic macromolecular protein assemblies through which mechanical force and regulatory signals are transmitted between the extracellular matrix (ECM) and the intracellular actin cytoskeleton:

- Transmembrane Core: Integrin alpha-beta heterodimers (e.g., alpha-5-beta-1 binding fibronectin).
- Core Structural Adaptors: Talin and Vinculin directly linking integrin beta-tails to F-actin.
- Enzymatic & Signaling Kinases: Focal Adhesion Kinase (FAK), Src kinase, Paxillin.

2. Mechanotransduction Mechanism (Talin-Vinculin Spring):

- Intracellular actomyosin contractility (mediated by non-muscle myosin II pulling on actin stress fibers) exerts mechanical tension on talin.
- Mechanical stretching of talin unfolds its hidden rod domains, exposing cryptic binding sites for Vinculin.
- Vinculin recruitment stabilizes the linkage, crosslinking additional actin filaments into thick, bundled contractile stress fibers.
- Concurrently, mechanical stretch induces FAK auto-phosphorylation at Tyr-397, creating a high-affinity binding site for the SH2 domain of Src kinase.
- Active FAK-Src complex phosphorylates downstream substrates (paxillin, p130Cas), activating Rac1 and RhoA signaling pathways that govern cell spreading, survival (suppression of anoikis), and directional durotaxis (migration toward stiffer substrate matrices).

>>Solution to Question 18: Describe the mechanism of Ewald curvature for the Transcytosis Establishment of TGN into the plasma membrane via Constitutive versus R

- Universally present in all eukaryotic cells.
- Operates continuously without requiring extracellular stimulation.
- Functions: Supplies newly synthesized proteins and lipids to the plasma membrane (e.g., receptors, nutrient transporters) and continuously secretes extracellular matrix components (e.g., collagen, fibronectin, serum albumin).
- Sorting: Proteins lacking specific targeting signals enter this pathway by default in non-clathrin-coated carrier vesicles that bud from the TGN and fuse spontaneously with the plasma membrane.

2. The Regulated Secretory Pathway (Signal-Mediated Secretion):
- Specialized in secretory cells (e.g., pancreatic beta-cells, endocrine cells, neurons, mast cells).
- Cargo Sorting and Condensation: Secretory products (e.g., insulin, neurotransmitters, digestive zymogens) possess specific structural patches that drive aggregation in the trans-Golgi in the presence of acidic pH (6.2) and high [Ca2+] (10 mM). Aggregated cargo is packaged into immature clathrin-coated secretory granules.
- Maturation: Proton pumps acidify the granule lumen, activating endoproteases (PC1/PC2, carboxypeptidase E) that cleave prohormones (e.g., proinsulin to mature insulin + C-peptide). Water is extruded, producing a highly concentrated dense-core secretory granule.
- Regulated Fusion: Granules dock and prime at the plasma membrane, arrested by complexin clamps. Fusion occurs only when an extracellular stimulus (e.g., glucose, action potential) triggers a rapid surge in cytosolic [Ca2+], binding synaptotagmin to release the clamp and fire exocytosis.

>> Solution to Question 28

>> Solution to Question 29 Explain the functional role of the Retromer complex in endosome-to-Golgi retrograde transport and its clinical significance in Alzheimer's

[Problem Statement]: Detail the molecular mechanism of Caveolae formation and their physiological functions in endothelial mechanosensing an
[Detailed Step-by-Step Answer]:
Caveolae Ultrastructure, Biogenesis, and Functions:
1. Structural Architecture:
Caveolae ('little caves') are flask-shaped, 50-100 nm invaginations of the plasma membrane abundant in endothelial cells, adipocytes, and smooth muscle:
- Caveolins (Cav1, Cav2, Cav3): Integral hairpin membrane proteins with both N- and C-termini facing the cytoplasm and a central hydrophobic intramembrane domain inserting into the inner leaflet.
- Cavins (Cavin-1 to Cavin-4): Soluble peripheral proteins oligomerizing into a striated polyhedral coat over the bulbous caveolin membrane surface. Cavin-1 (PTRF) is strictly essential for caveolar morphology.
- Lipid Composition: Highly enriched in cholesterol and sphingomyelin, forming a specialized, immobile liquid-ordered microdomain.
2. Physiological Functions:
- Endothelial Transcytosis: Facilitates non-clathrin-dependent uptake and transport of macromolecules (serum albumin, LDL, insulin) across vascular endothelial barriers via Cav1-dependent budding.
- Mechanical Buffer / Shear Stress Reservoir: In response to sudden hemodynamic pressure or mechanical cell stretch, caveolae rapidly flatten into the plane of the plasma membrane without breaking, buffering membrane tension and preventing membrane rupture.
- Signal Transduction Platform: Scaffolds endothelial nitric oxide synthase (eNOS), receptor tyrosine kinases, and G-protein subunits within its scaffolding domain (CSD), regulating nitric oxide production in response to laminar blood flow.

>> **Solution to Question 22:** [Problem Statement]: Explain how bacterial pathogens like Listeria and Shigella utilize host cell actin polymerization for intracellular motility and cell-to-cell sp

>>Solution Question 24: Describe the structure and role of the Nuclear Envelope. How does it connect to the cytoskeleton?

mechanical bridge.
3. Physiological Functions:
Transmits extracellular mechanical forces from focal adhesions directly to the genome, regulating nuclear positioning, chromosome movement during meiosis, and mechanosensitive gene expression.

The nascent phagosome undergoes a series of hit-and-run fusion events with endocytic compartments:
- Early Phagosome: Acquires Rab5, EEA1, and V-ATPase, dropping pH to ~6.0.
- Late Phagosome: Rab conversion replaces Rab5 with Rab7, recruiting LAMP-1 and acid hydrolases.
- Phagolysosome: Fuses with lysosomes. Luminal pH plummets to ~4.5. Assembly of the phagocyte NADPH oxidase complex (NOX2) produces massive bursts of superoxide radicals (O_2^-), while myeloperoxidase generates hypochlorous acid (HOCl), killing the engulfed pathogen.

>>Solution Question 28: Describe the role of Spectrin and associated cytoskeletal proteins in maintaining the biconcave shape and mechanical stability of the erythrocyte membrane.

- Membrane Diffusion Barriers: Form tight collar rings at constricted cellular necks, restricting lateral diffusion of membrane proteins:
- At the cleavage furrow during cytokinesis, preventing premature mixing of dividing cells.
- At the base of the primary cilium, maintaining distinct ciliary membrane protein composition.
- At the axon initial segment (AIS) of neurons, maintaining axonal versus somatodendritic polarization.
- Scaffolding Machinery: Act as a physical platform recruiting actin-binding proteins, exocyst tethering complexes, and signaling kinases during cell division and phagocytosis.

- Activates a cryptic splice donor site 150 bp upstream, yielding an mRNA deleted for 150 nucleotides.
- The resulting mutant protein, Progerin, has an internal deletion of 50 amino acids that removes the ZMPSTE24 endoproteolytic cleavage site!

3. Cellular Pathology:

Progerin undergoes farnesylation and carboxymethylation normally, but cannot undergo the final cleavage step. Progerin remains permanently farnesylated and membrane-bound, accumulating abnormally at the inner nuclear membrane. This causes:

- Severe lobulation and nuclear envelope blebbing.
- Loss of peripheral heterochromatin and histone H3K9me3 marks.
- Disruption of the LINC complex, accumulation of double-strand DNA breaks, telomere attrition, and premature cellular senescence.

4. Mechanism of Farnesyltransferase Inhibitors (FTIs, e.g., Lonafarnib):

- Lonafarnib competitively inhibits the protein farnesyltransferase enzyme, blocking attachment of the farnesyl lipid tail to mutant prelamin A.
- Without the hydrophobic anchor, unfarnesylated progerin cannot remain trapped at the inner nuclear membrane.
- FTI treatment significantly restores normal nuclear morphology, reduces DNA damage signaling, improves cardiovascular arterial compliance, and extends survival in HGPS clinical trials.
